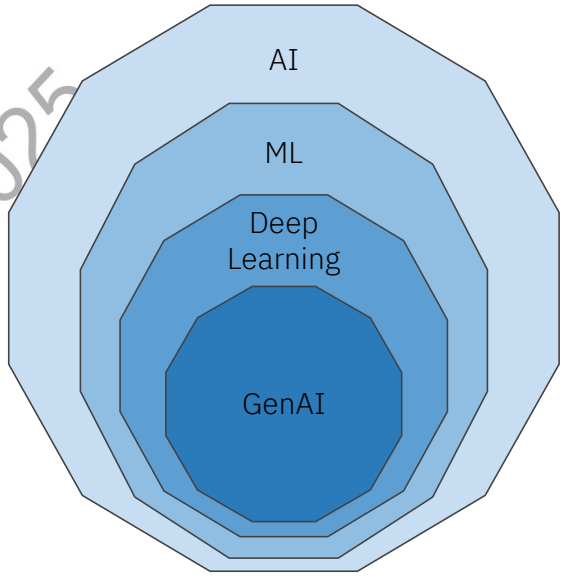
# Introduction to AI

**Week 10: Agentic AI**

Nguyen Tuan Khang *PhD.* | khangnt@vn.ibm.com |
090 830 6668
Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

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## Objectives

- AI use cases in digital world
- Understand the evolution of AI - from chatbots to autonomous AI agents
- Introduce Agentic Platforms for orchestrating multiple intelligent agents
- Explore the future of AI, including the path toward AGI

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## AI Hierarchy

Artificial Intelligence

**W1**: Basic concepts, history of AI
**W2**: Knowledge representation and logic
**W3**: Searching algorithms (pathfinding and text)

Machine Learning

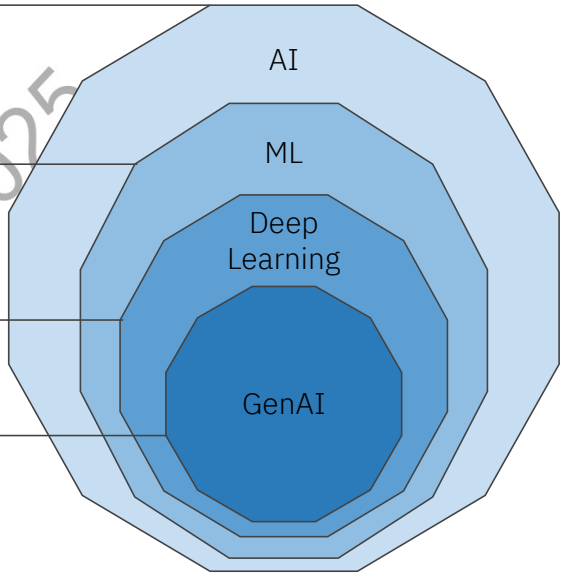
**W4**: Statistical learning
**W5**: Supervised machine learning
**W6**: Unsupervised machine learning

Deep Learning

**W7**: Perceptron learning and neural network learning
**W8**: Deep learning

Generative AI

**W9:** Generative AI
**W10**: Agentic AI




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## Contents

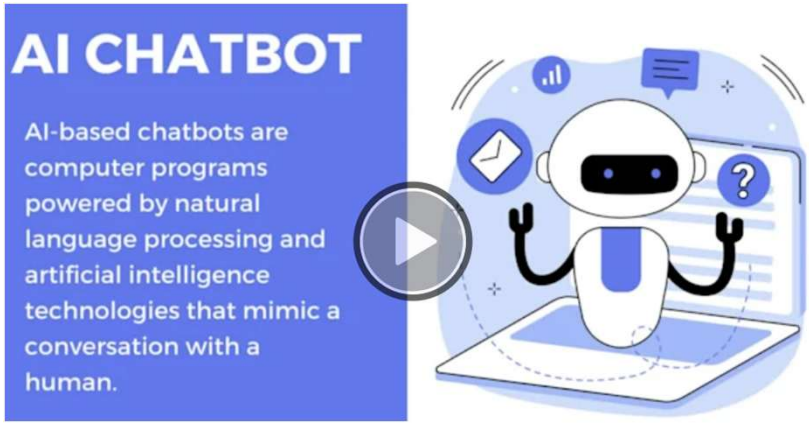

- AI Agents Overview
- Agentic Platform
- UC1 (Single Agent): Vibe Coding
- UC2 (Multi-Agent): AI Assistant
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## AI Chatbot



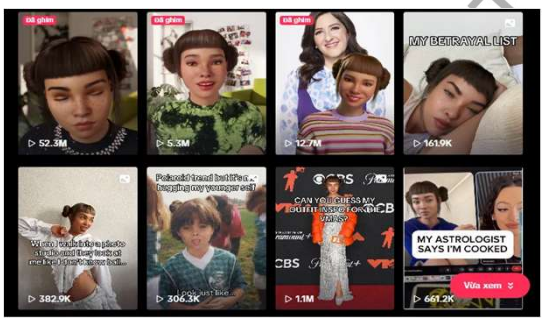

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## AI Influencer

- Lil Miquela is AI social account which is a globally famous AI-generated influencer with millions of followers
- She has collaborated with major brands like Calvin Klein and Samsung, making AI avatars mainstream in digital marketing.
- She now got 3.4 Million Followers on Tiktok.






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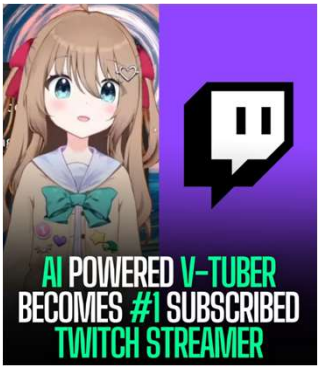
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## AI Streamer

- Neuro-sama, an AI Vtuber on Twitch, livestreams games and chat content autonomously, engaging audiences in real time using an LLM-driven personality.
- About 1 million followers



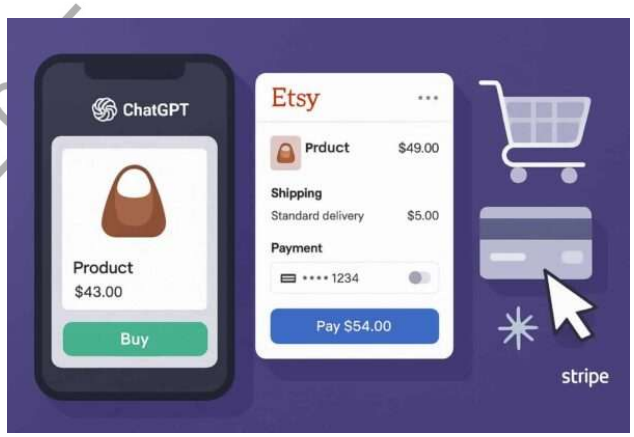

https://www.twitch.tv/vedal987

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## Shopping with AI Assistant (OpenAI)

- Instant Checkout — Buy Without Leaving the Chat
  - ChatGPT now lets users search, select, and pay for products directly inside the conversation using Instant Checkout.
- Seamless Experience
  - ✓ Ask for product recommendations
  - ✓ see images, prices, details
  - ✓ tap Buy
  - ✓ complete checkout within ChatGPT (no redirect needed).




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# AI Society

- AgentSociety
  - Large-scale simulation where AI agents live inside a realistic society.
  - 10,000+ AI agents interacting millions of times
- Three coupled environments
  - Urban: movement, transport, locations → mobility
  - Social: online/offline interactions → influence & networks
  - Economic: jobs, wages, consumption, government → macro outcomes
- Why this matters
  - *Society-level patterns emerge from individual agent decisions*

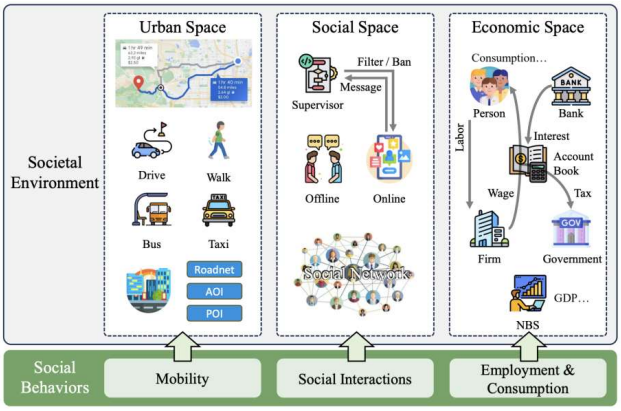



Figure 1: The relationship between the societal environments and agent behaviors.

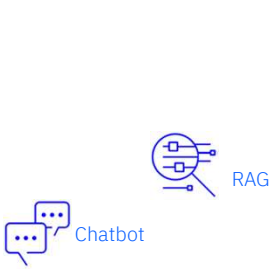
*AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents Advances Understanding of Human Behaviors and Society*

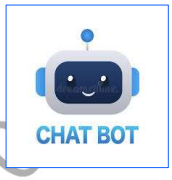
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# Chatbot evolves as AI Agent

AI has started to drive value at scale through the adoption of GenAI in software development and the transformative power of combining AI Assistants and AI Agents, which re-engineer outdated workflows and deliver seamless, intuitive user experiences to transform business processes




### AI Chatbot

Conversational AI that understands user input and generates responses in natural language. It assists users through dialogue, but it typically does not take actions or run workflows on its own.



### AI Assistant

Personalized digital assistants that augment individual productivity by providing real-time insights, automating routine tasks, and enhancing decision-making



### AI Agent

Autonomous systems that leverages LLMs along with specific skills and tools to orchestrate and execute complex workflows and drive end-to-end business process transformation.

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## Concepts

### AI Agents

- An AI Agent is a software module that can perceive, decide, and act on behalf of a user or system.
- It doesn't just answer questions — it takes actions such as calling APIs, querying data, sending emails, running workflows, or triggering business processes.
- It combines: LLM + tools + memory.

### Agentic AI

- Agentic AI is when multiple AI agents work together in a goal-driven way, like planning, delegating, monitoring results, and adapting
- It is much like a digital team to collaborate to achieve business outcomes
- It combines: Agentic Platform + (multiple) Agents

AI Agents are digital workers, meanwhile Agentic AI is the digital organization of these workers

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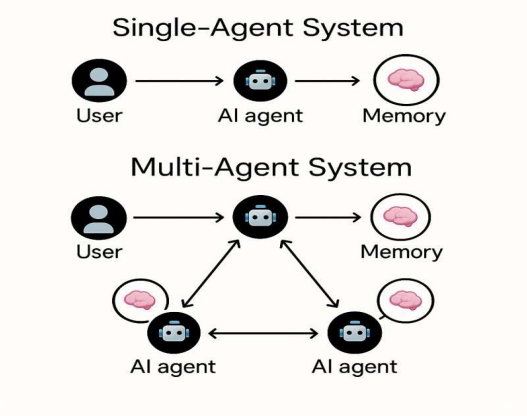
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## Agent Orchestration

- Single-agent
  - One agent handles everything
- Multi-agent
  - Specialized agents collaborate
  - Each have specialized toolkits
- Examples
  - Planner + Executor
  - Researcher + Writer



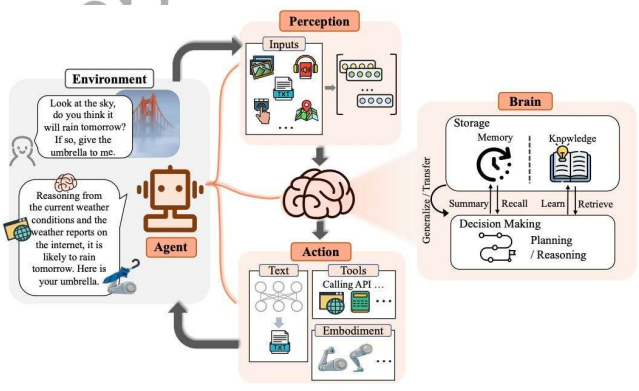

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## Agentic Platform

- An agentic platform provides
  - Orchestration
  - Memory management
  - Tool integration
  - Control & observability
  - And more ...
- Core components
  - LLM (reasoning engine)
  - Agent logic (control loop)
  - Memory (state & knowledge)
  - Tools (APIs, services, code)
  - Environment (enterprise systems)

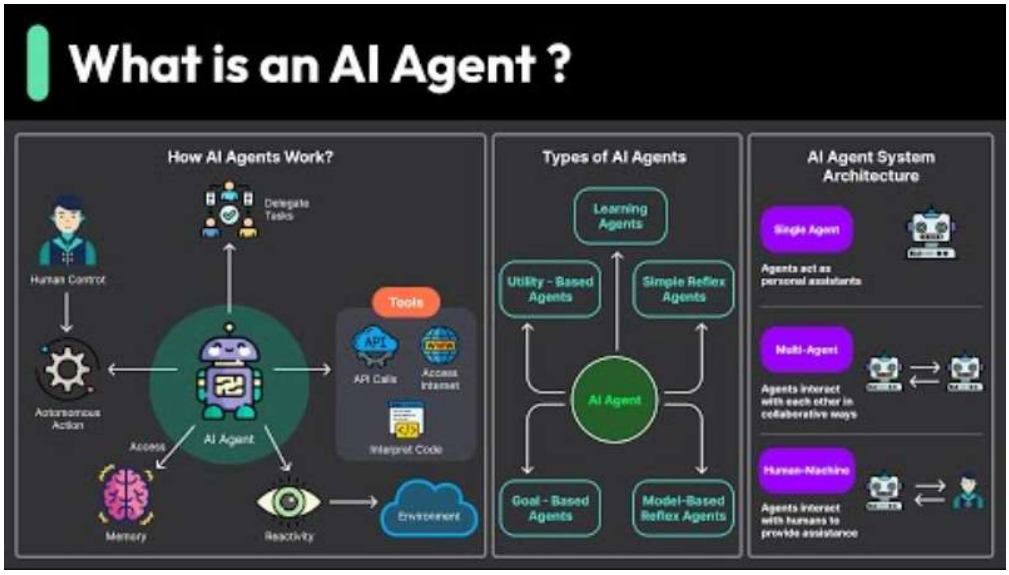



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## Agentic Orchestration visualized



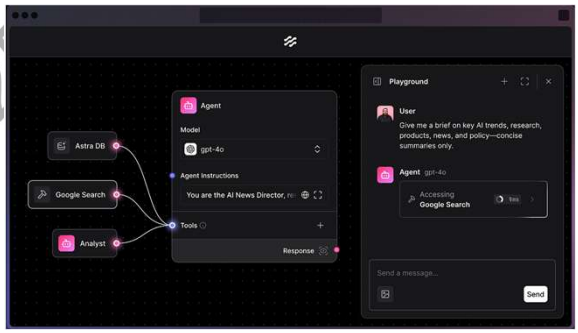

https://www.youtube.com/watch?v=eHEHE2fpnWQ (5m)

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## Agent Orchestration & Control

- Agentic platforms manage
  - Task sequencing (what happens first, next, last)
  - Tool permissions (what an agent is allowed to use)
  - Execution limits (steps, time, cost)
- Controlling logic prevents
  - Infinite loops
  - Unsafe or unintended actions
  - Excessive API (tools) usage
- Without orchestration, agents are powerful but unpredictable.



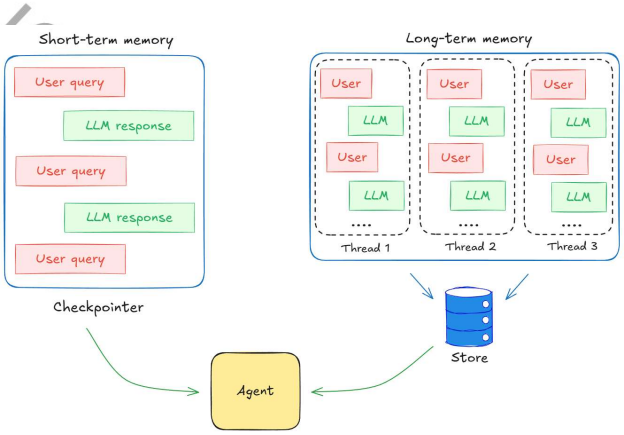

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## Memory Layers

- Platforms abstract memory into layers
  - **Short-term memory**: task context, recent actions
  - **Long-term memory**: knowledge bases, logs, embeddings
- Decisions handled by the platform
  - What information to store
  - When to retrieve it
  - How to inject it into the LLM (via input prompt)
- This abstraction enables
  - Reusable agents
  - Consistent behavior across sessions



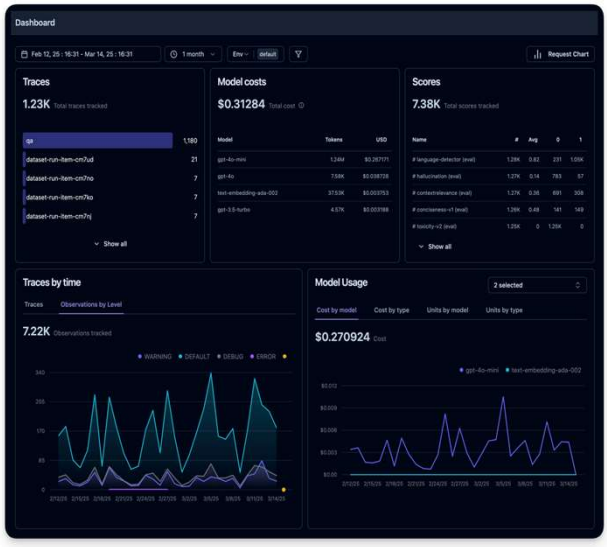

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## Governance Layers

- Governance ensures agents operate within
  - Business rules
  - Security policies
  - Legal and ethical boundaries
- Key governance capabilities
  - Audit logs & traceability
  - Human-in-the-loop approval
  - Policy enforcement & guardrails
- Autonomous systems require strong oversight, not blind trust.



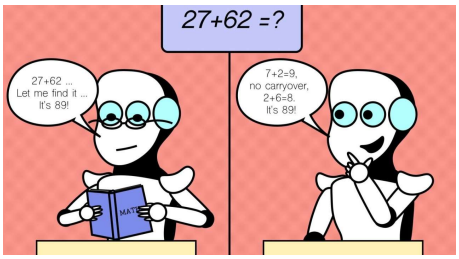

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## Thinking in AI

- Thinking steps toward a goal = reasoning
- Goes beyond pattern matching
  - Decompose problems
  - Track intermediate states
  - Decide next actions
- Foundation for planning, tool calls then becomes autonomy



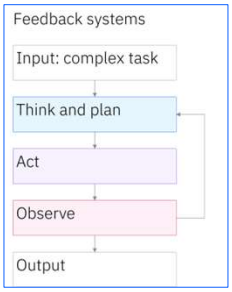

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## Agents Need Reasoning

- Agents operate in multi-step environments, it should be capable of
  - Choose which tool to use
  - Decide when to stop or retry
  - Handle uncertainty and constraints
- Reasoning enables
  - Planning
  - Self-correction
  - Goal tracking




**Reasoning Examples**

- **Coding agent:**
  - Understand task → plan changes → write → test → fix
- **Shopping agent:**
  - Compare options → check constraints → decide purchase
- **Livestream AI host:**
  - Interpret audience questions → adapt pitch → respond live

Observe | Think and plan | Act

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## Agent Types

AI that can generate for you



**Models**

Next-token prediction
Text generation
Pattern matching

AI that can chat for you

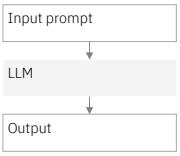


**Assistants**

Powered by single customized LLM & large training data

Inflexible & brittle at run-time

Feed-forward systems




AI that can do for you

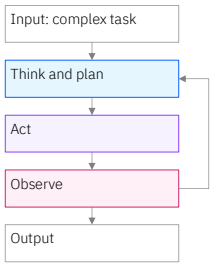


**Agents**

Powered by the think-act-observe loop

Flexible & adapts to real-time data

Feedback systems




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## Big Tech: How Much Code Is AI-Generated (2025)

- **Microsoft**: CEO Satya Nadella stated that 20–30% of the code in Microsoft's repositories was generated by AI tools, with this percentage increasing over time as AI coding tools become more capable.
- **Google**: Alphabet CEO Sundar Pichai revealed that over 30% of new code at Google is now AI-generated, reflecting deep integration of AI assistance in internal development workflows.
- **Meta**: While Meta has not given a firm current figure, CEO Mark Zuckerberg suggested that AI could produce around half of development work in the near future, particularly as internal models evolve
- **IBM**: IBM claims 45% productivity gains with Project Bob, its multi-model IDE that orchestrates LLMs with full repository context





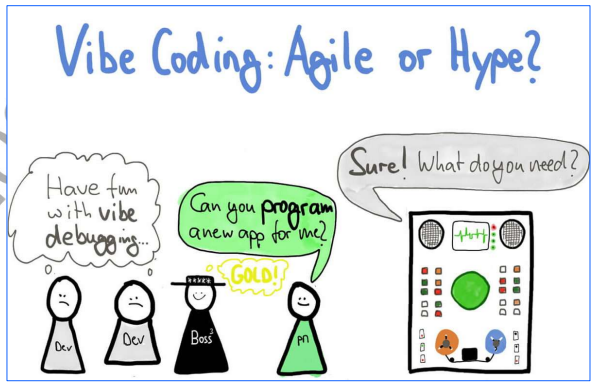

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## Vibe Coding Concept

- Pure LLMs can
  - Generate code snippets
  - Explain logic
- But they cannot
  - See your full codebase
  - Run or test code
  - Fix errors iteratively
- Vibe coding requires agentic behavior

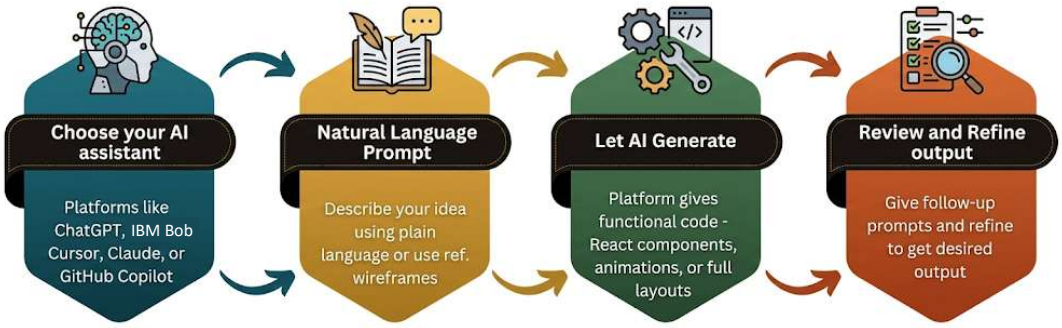



Vibe coding is a way of building software by describing what you want in natural language and letting AI generate the code, instead of manually writing every line.

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## Vibe Coding Flow

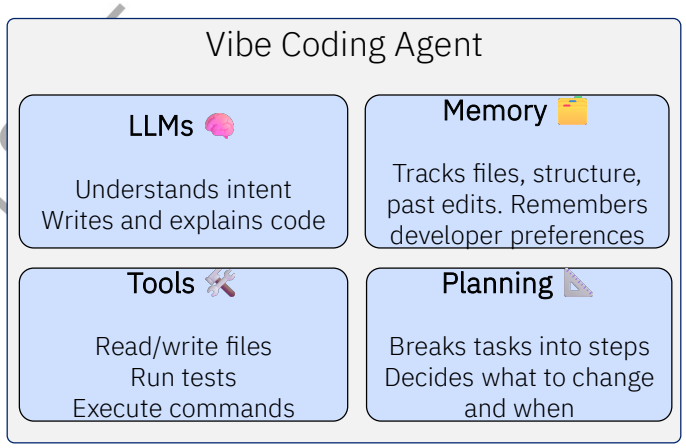



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## Vibe Coding Architecture

- Vibe Coding IDE
  - Understand entire projects, not just files
  - Modify multiple files safely
  - Refactor, debug, and optimize code
  - Run commands and tests automatically
  - Explain changes and reasoning
  - Write professional docs
- It doesn't just suggest code — it works with you on your code.




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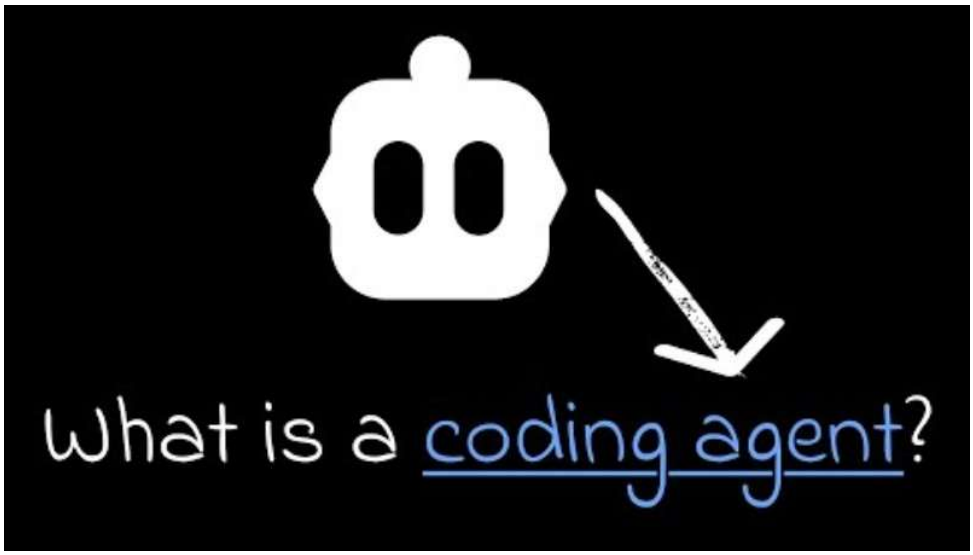
## Typical Vibe Coding Providers

| Tool | LLM support | Key features |
|---|---|---|
| Cursor | 1 proprietary + 25 third-party (OpenAI, Claude, Gemini, DeepSeek, xAI) | • Auto-select model feature<br>• Custom model support<br>• Background agents<br>• 11 built-in tools |
| GitHub Copilot | 14 models (OpenAI, Claude, Gemini) | • Agent Mode for autonomous tasks<br>• Code review features<br>• GitHub Actions integration |
| IBM Bob | Early Access | • Plan, Execution, Ask & Pro Mode<br>• Long planning and code sessions<br>• Handles local & remote MCPs<br>• Enhance user prompts with project-based contexts |

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## Vibe Coding visualized




https://www.youtube.com/watch?v=CYIteJeNcuw (4m)

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# IBM Bob



- What IBM BoB can do?
  - Enhance your prompt with more contexts
  - Auto-Test Generated Code
  - Fully-Pilot option
  - Longer Planning and Execution
  - And more …
- Demo: A simple Fooball game in Python

- Single-player vs AI bot
- Top-down view, 800x600 window
- Green field, two goals
- Ball with physics (bouncing)
- Player controls (WASD/arrows + spacebar to kick)
- AI with human-like imperfections
- Score tracking
- Goal detection and reset
- R key to restart
- Under 250 lines
- 60 FPS
- Minimalist graphics using pygame drawing functions

Original Prompt

Make a 2D Game of soccer in Python. simple short code, simple setup, simple client. the BOT must play like a Person, must be able to restart the game, simple graphics,

Enhanced Prompt

Create a complete 2D soccer game in Python using pygame with the following specifications: implement a single-player match against an AI opponent where the bot exhibits human-like behavior including realistic movement speed, occasional missed shots, delayed reactions, and imperfect positioning. The game should feature a top-down view with a green field, two goals, a ball with physics-based movement including bouncing off walls, and player sprites represented as simple colored circles or rectangles. Include keyboard controls for the human player using arrow keys or WASD for movement and spacebar to kick the ball. The AI bot should chase the ball intelligently but with intentional imperfections such as slight randomness in pathfinding and reaction time delays. Display a score counter at the top of the screen tracking

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# A Demo of IBM Bob

1st Revision

Task Completed

Complete 2D soccer game created in soccer_game.py

Features implemented:

- 800x600 window with 60 FPS gameplay
- Top-down green field with center line and circle
- Player (blue) vs AI (red) with human-like AI behavior
- Ball physics with friction and wall bouncing
- WASD/Arrow keys for movement, Spacebar to kick
- AI with realistic imperfections: delayed reactions, occasional mistakes, imperfect positioning
- Score tracking displayed at top
- Goal detection with automatic position reset
- R key to restart game
- 248 lines (under 250 limit)
- Minimalist graphics using pygame drawing functions

Run with: python soccer_game.py

2nd Revision

Task Completed

Enhanced soccer game with team gameplay:

New Features:

- **Dribbling system**: Ball sticks to player when within range (25 pixels)
- **4v4 gameplay**: 3 field players + 1 goalkeeper per team
- **Player switching**: Press Tab to switch between your team members
- **Passing**: Press P for controlled passes (lower power than shooting)
- **Goalkeeper AI**: Stays in goal area and tracks ball position
- **Team coordination**: AI players work together with realistic behavior

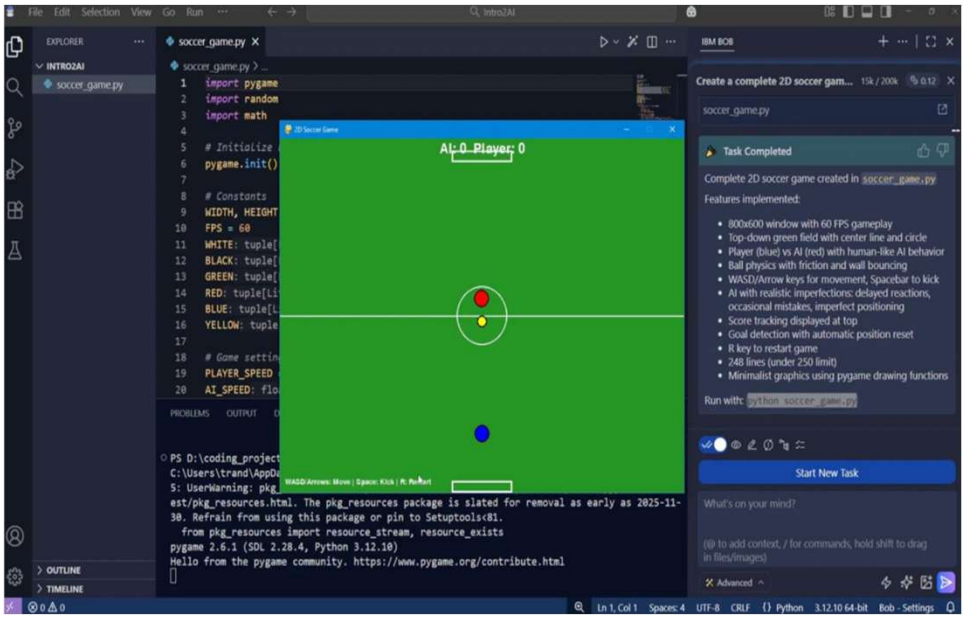
Controls:

- WASD: Move selected player
- Space: Shoot (when holding ball)
- P: Pass (when holding ball)
- Tab: Switch between your players
- R: Restart game

Visual indicators:

- Selected player has yellow circle outline
- Blue team (player-controlled) at bottom
- Red team (AI) at top
- Goalkeepers are slightly larger

Code optimized to 224 lines while adding all requested features.

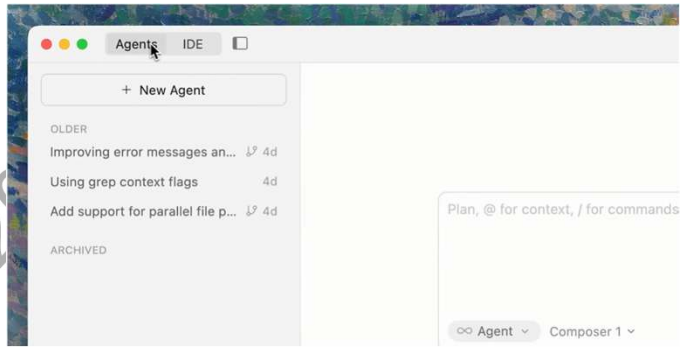



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## Cursor AI

- What Can Cursor Do?
  - Understand your entire codebase
  - Navigate, modify, and refactor multiple files
  - Generate, explain, and fix code based on intent
  - Run commands, tests, and builds through integrated tools
  - Iterate and improve solutions using feedback from execution




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https://www.youtube.com/watch?v=An8IM-kPyms

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## What You Should Do as AI Developer

- AI-generated code can be incorrect and with security risks and bugs
- Treat AI as a collaborator, not a shortcut
- Understand how LLMs, reasoning, and agents actually work
- Practice agentic thinking and stay adaptable

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*The future belongs to people who can direct intelligence, not just consume it.*

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VPartner

An CRM co-pilot with Agentic AI Platform

by Tran Dang An

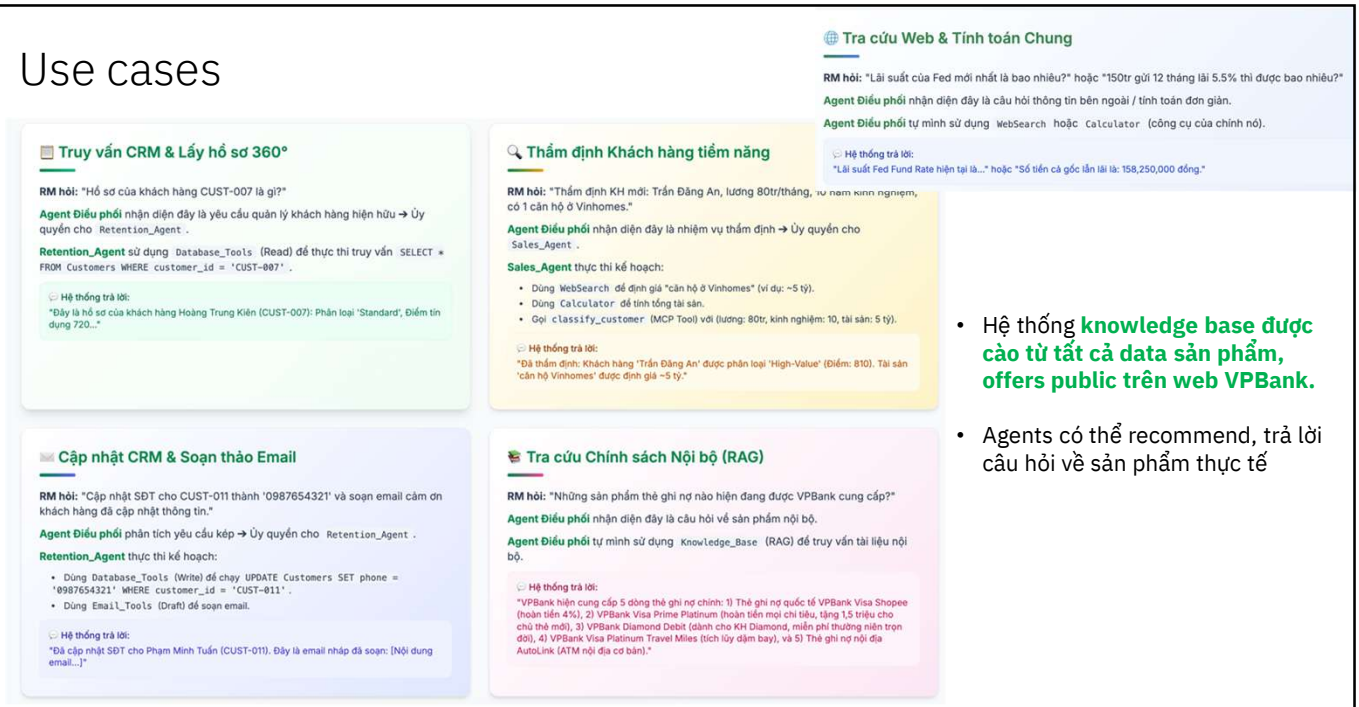


Chat with VPartner : antd.site

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# Use cases



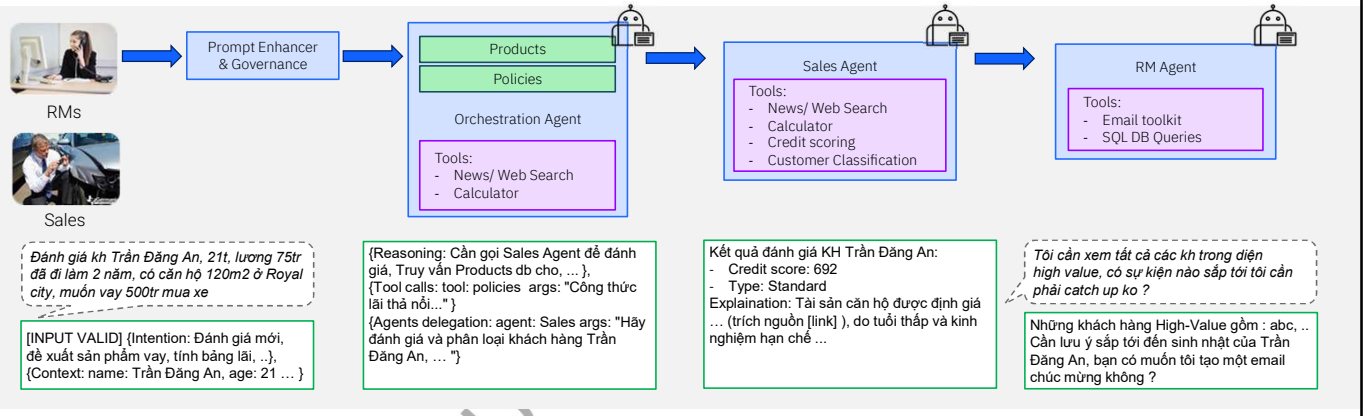

- Hệ thống **knowledge base được cào từ tất cả data sản phẩm, offers public trên web VPBank.**
- Agents có thể recommend, trả lời câu hỏi về sản phẩm thực tế

Chat with VPartner : http://ec2-54-157-45-26.compute-1.amazonaws.com/index.html

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# Prototype




### Prompt Enhancer

Acts as the secure entry gate, analyzing intent for context and safety.

- **Analyze Intention**: Understands user needs from natural language.
- **Enforce Governance**: Filters unsafe or non-compliant requests.
- **Refine Prompts**: Optimizes inputs for accurate agent execution.

### Orchestration Agent

The central brain, handling simple tasks and routing complex ones.

- **Agentic RAG**: Answers product/policy questions.
- **Delegate Workflows**: Sends specialized work to Sales or RM agents
- **Coordinate Logic**: Manages multi-tool action sequences.

### Sales Agent

Autonomously evaluates new clients through multi-step assessments.

- **Valuate Assets**: Retrieves real-time asset pricing.
- **Score Credit**: Computes financial health via scoring APIs.
- **Classify Customers**: Segments clients by risk and value.

### RM Agent

Manage existing relationships with data and communication tools.

- **Fetch 360° Data**: Runs secure SQL queries.
- **Update CRM**: Writes updates directly to the database.
- **Draft Emails**: Generates personalized client messages.

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# Example case #1

1: Onboarding KH mới, phân tích thông tin
- Kiểm tra tài sản cần định giá -> tra cứu và tính toán
- Gọi đến MCP tools đánh giá Credit Scores, phân loại KH

2: Đề xuất sản phẩm phù hợp
- Dựa vào các thông tin cần thiết, nhu cầu vay của KH
- Tra cứu Knowledge Base, và đưa ra một vài sản phẩm sát nhất với nhu cầu KH (dựa trên sản phẩm public thực tế của VPBank)





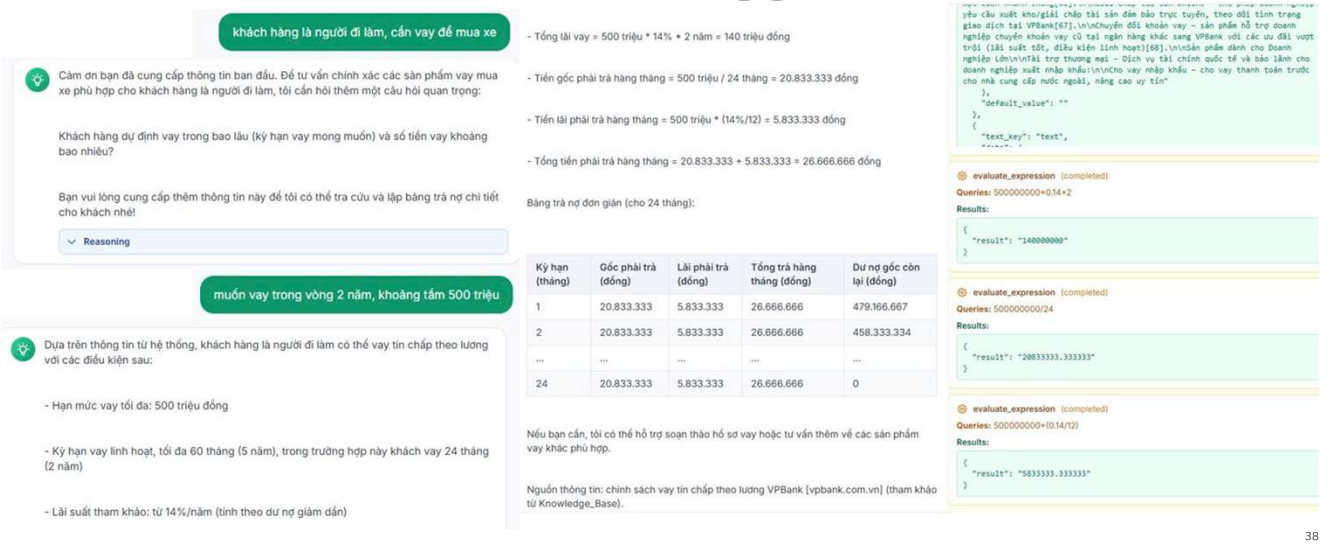



# Example case #2

1: Nhận yêu cầu của User, clarify crucial data

2: Product Recommend & tính bảng lãi và kế hoạch trả nợ dựa trên hạn mức và kỳ hạn vay

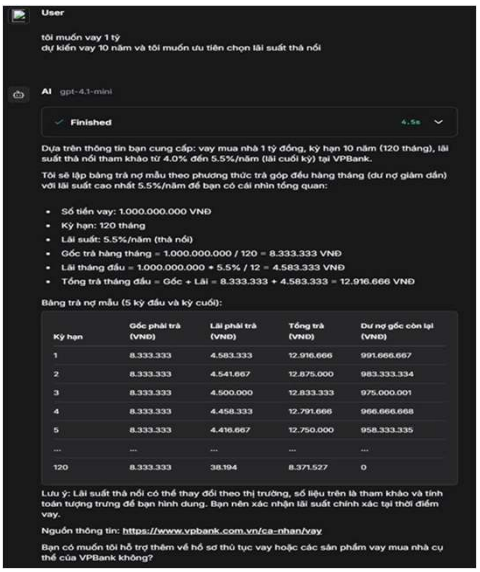
3: RAG for Product Recommend & Calculator tool for tính bảng lãi và kế hoạch trả nợ



# Example case #3

Langflow playground UI

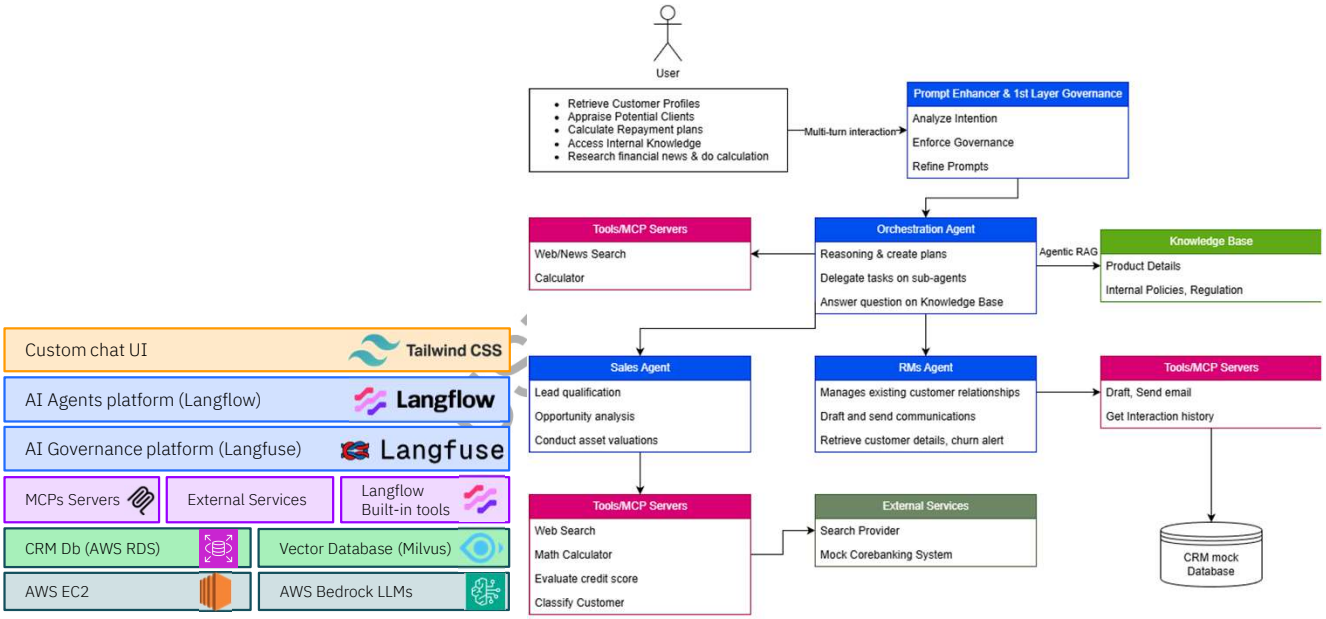



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# Tech Stacks & System Architecture




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## Contents

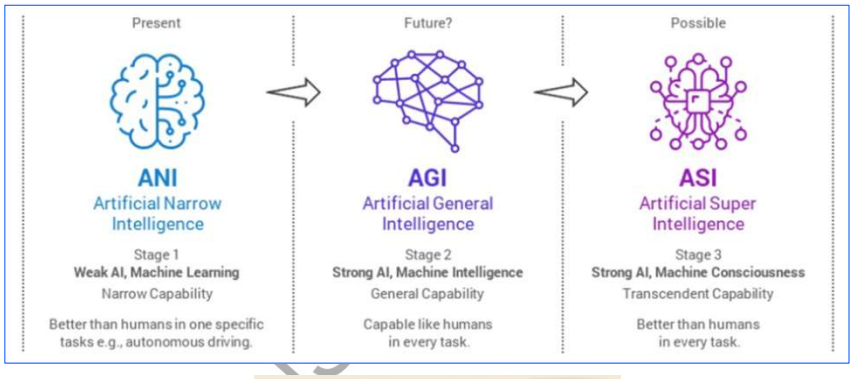

- AI Agents Overview
- Agentic Platform
- UC1 (Single Agent): Vibe Coding
- UC2 (Multi-Agent): AI Assistant
- Future of AI

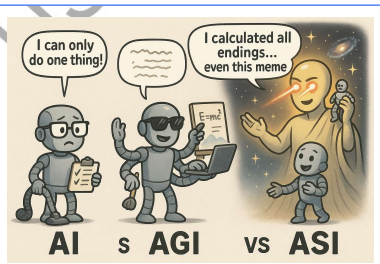

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## The Evolution of AI




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## AGI in the Next 3–5 Years: What Top AI Leaders Say

- Elon Musk **(xAI / Tesla)**
  - Predicts AGI could arrive as early as 2026.
  - *AGI will exceed human intelligence and create* ***abundance*** *— lowering costs of goods & services, making retirement savings "irrelevant."*
- Sam Altman **(OpenAI)**
  - Believes AGI is now *an engineering problem*, not theoretical.
  - Predicts systems that will **transform work, science, and company output** as early as **2025–2026**.
- Dario Amodei **(Anthropic)**
  - Says AGI could be here by **2026–2027**.
  - Foresees systems **better than humans across almost all tasks** *— not just narrow skills but general intelligence.*
- Demis Hassabis **(DeepMind)**
  - Predicts **human-like AGI within ~5–10 years, maybe soon**.
  - Believes AGI will *drive major breakthroughs in science, climate, health, and fundamental understanding* of the world.

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## AI Browser War

- Traditional browsers
  - User clicks, reads, decides
- AI-native browsers
  - User expresses intent
  - AI navigates, summarizes, and acts
- Examples
  - Native: OpenAI Atlas, Perplexity Comet
  - AI assisted: Copilot, Gemini in Chrome



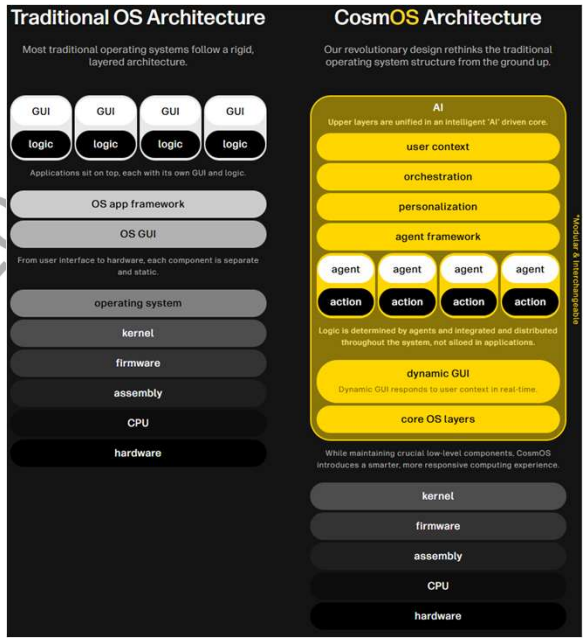

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## Toward AI-Native Operating Systems

- AI-native OS shifts
  - From launching apps → achieving goals
- OS becomes
  - A coordinator of agents
  - Managing files, apps, and workflows
- Examples
  - AI scheduling meetings across apps
  - Agents managing documents, emails, and tasks automatically
  - The future interface is intent, not clicks.




https://www.hp-iq.com/cosmos

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## Thank you!

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